ROVER POWER DISTRIBUTION BOARD

6S battery -> shunt -> three LMR51450 converters

11.99 V: Jetson | 4.99 V: hobby servos | 22.19 V: DYNAMIXEL

Battery input and INA226

RS1 + U1

Battery voltage / current

6S: (19.8, 22.2, 25.2) V

Planning range; not a cutoff setting.

VBAT

I2C_SDA

I2C_SCL

File: input-monitor.kicad_sch

12 V Jetson supply

U2 | 11.989 V | 3 A load

VBAT

Jetson + 12 V expansion

File: buck-12v.kicad_sch

VBAT

5 V servo supply

U3 | 4.988 V | 2.4 A load

VBAT

Three servos + 5 V expansion

File: buck-5v.kicad_sch

Pico 2 and hobby servos

SKT1 + J2-J4

I2C_SDA

I2C_SCL

Pico: USB power/data; 4 GPIO expansion

File: pico-servos.kicad_sch

22.19 V motor supply

U4 | 22.190 V | 3.5 A load

VBAT

VMOTOR

Motor supply + SW1 + rail indicator

File: buck-motor.kicad_sch

DYNAMIXEL distribution

J1 -> J5/J6/J7

VMOTOR

Shared motor power and data bus

File: dxl-enable.kicad_sch

READING THIS SCHEMATIC

2 Battery + INA226

3 12 V converter

4 5 V converter

5 Motor converter

6 Pico + servos

7 DYNAMIXEL distribution

Loads are planning figures, not validated PCB ratings.

REVIEW HOLD: motor braking energy / output overshoot; external battery protection; servo pinout; DC-bias capacitance.

Pico uses Jetson USB power and data. Motor enable follows VBAT through a switch-safe resistor network.

REVISION NOTES | 2026-09-30

A1: integrated drawing. A2: LMR51450 converter circuit.

A9: VIN-derived motor enable; USB-only Pico; sourced optional GPIO header. Stop before PCB layout.

HUMAN REVIEW - not approved for fabrication.
Schematic only. Owner edits are authoritative.
KSRC Lunar Rover

Sheet: /
File: rover-pdb.kicad_sch

Title: ROVER PDB | System overview

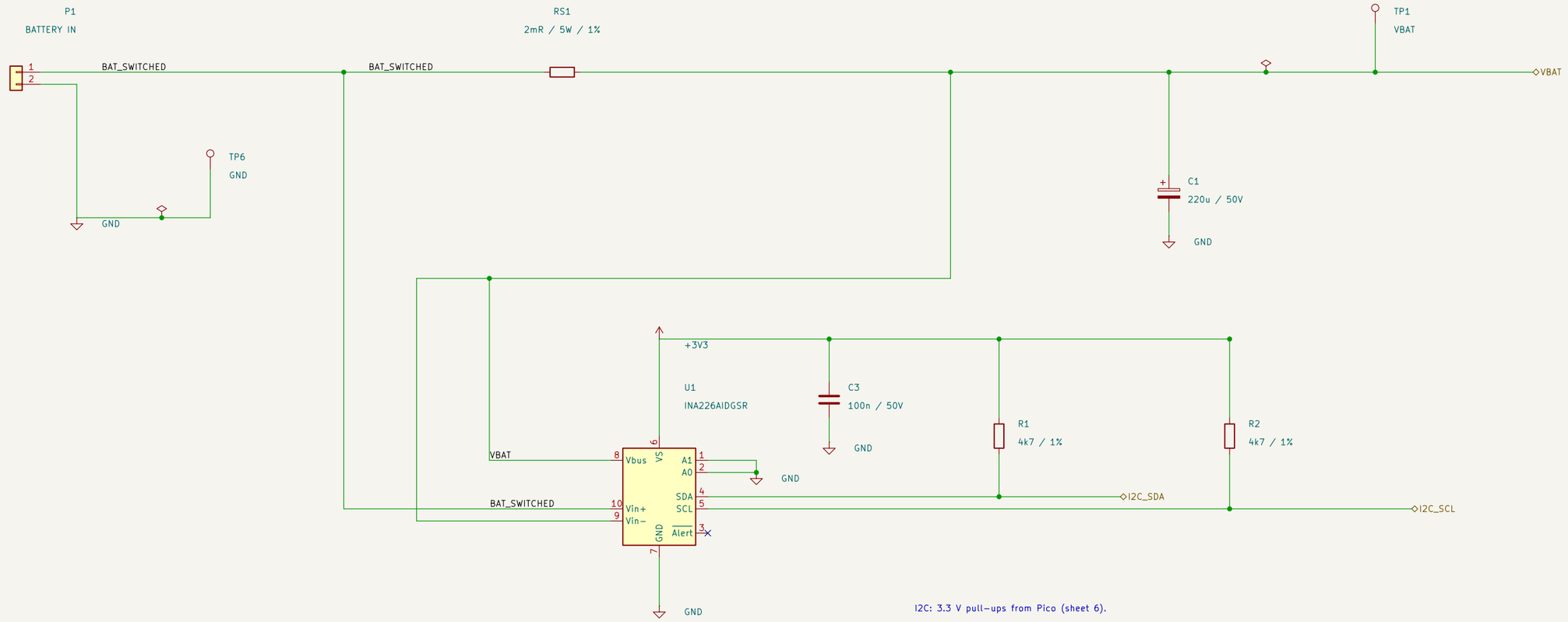
Size: A3	Date: 2026-10-04	**Rev: A10**
KiCad E.D.A. 10.0.6		Id: 1/7

BATTERY INPUT AND INA226

BATTERY INPUT / CURRENT AND VOLTAGE MEASUREMENT

External switch and XT60 are off-board. F1 removed by owner; no onboard battery fuse.

Battery: (19.8, 22.2, 25.2) V; positive discharge current flows left to right.



I2C: 3.3 V pull-ups from Pico (sheet 6).

VS is the 3.3 V supply; VBUS measures VBAT.

C3 bypasses VS, not the battery-sense pin.

C3 bypasses VS locally. R1/R2 are the shared I2C pull-ups. Named ports connect to the overview.

INA226: A1=A0=GND; 7-bit address 0x40 (1000000b). ALERT unused. Kelvin sense at RS1 terminals.

RS1: 10 A -> 20 mV, 0.20 W. I = Vshunt / 0.002 ohm. Current_LSB=1 mA; CAL=0x0A00; Power_LSB=25 mW.

No reverse-polarity protection, automatic battery cutoff, or input transient clamp in this draft.

P1 uses solid surface pads. Provide cable restraint; an external battery fuse remains recommended.

HUMAN REVIEW — not approved for fabrication.
Schematic only. Owner edits are authoritative.
KSRC Lunar Rover

Sheet: /Battery input and INA226/
File: input-monitor.kicad_sch

Title: ROVER PDB | Battery input and INA226

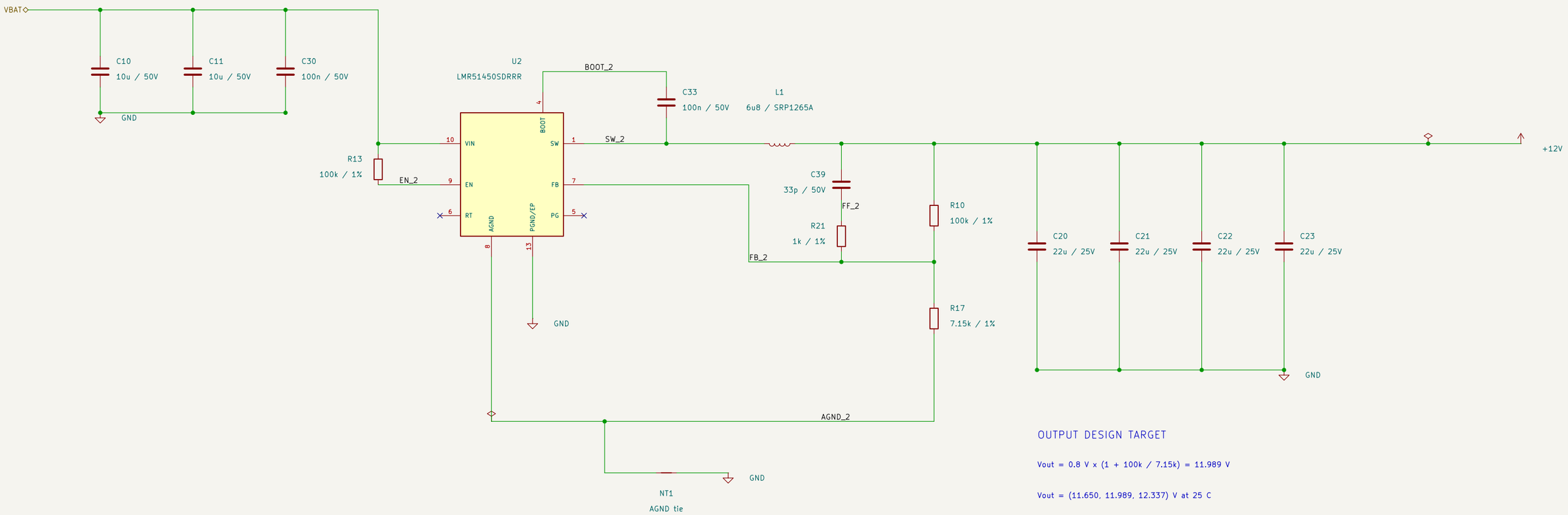
Size: A3	Date: 2026-10-04	**Rev: A10**
KiCad E.D.A. 10.0.6		Id: 2/7

12 V JETSON SUPPLY

VBAT = (19.8, 22.2, 25.2) V: planning battery range; no automatic cutoff.

INPUT AND ENABLE

REGULATOR / FILTER / FEEDBACK



OUTPUT DESIGN TARGET

Vout = 0.8 V x (1 + 100k / 7.15k) = 11.989 V

Vout = (11.650, 11.989, 12.337) V at 25 C

Bounds exclude ripple, transients and dropout.

Planning load: 3.0 A; board thermal rating unverified.

A6 Cout: 4 x 22u / 25V; about 35.8u typical at 12V.

TI Table 9-2: 20u effective; this bank includes margin.

CONFIGURATION AND IMPLEMENTATION

LMR51450SDRRR / C6890985: 5 A IC, PFM + spread spectrum; RT open = 500 kHz; PG unused.

Input HF capacitor and BOOT capacitor: short loops at IC. Join AGND to PGND/EP13 locally through the copper net tie.

Stacked package pins: VIN 10-12; SW 1-3. Solder EP13. Feed-forward RC and effective output capacitance require validation.

EN is pulled to VBAT. No battery undervoltage cutoff. Verify loaded startup, load steps and temperature rise.

Source: TI SLUSEP7, sections 8-9 and DRR drawing; www.ti.com/lit/ds/symlink/lmr51450.pdf

A6: reference-based capacitor count with margin. C20-C23: Samsung CL32B226KAJNNNE / C309062.

35.8u is typical after DC-bias loss, not a guaranteed minimum. C42/C43 removed; L1 stays 6.8uH.

Bench goals: 0.5% ripple / 5% load-step deviation. Verify actual load steps and startup; compliance not established.

HUMAN REVIEW — not approved for fabrication.
Schematic only. Owner edits are authoritative.
KSRC Lunar Rover

Sheet: /12 V Jetson supply/
File: buck-12v.kicad_sch

Title: ROVER PDB | 12 V Jetson supply

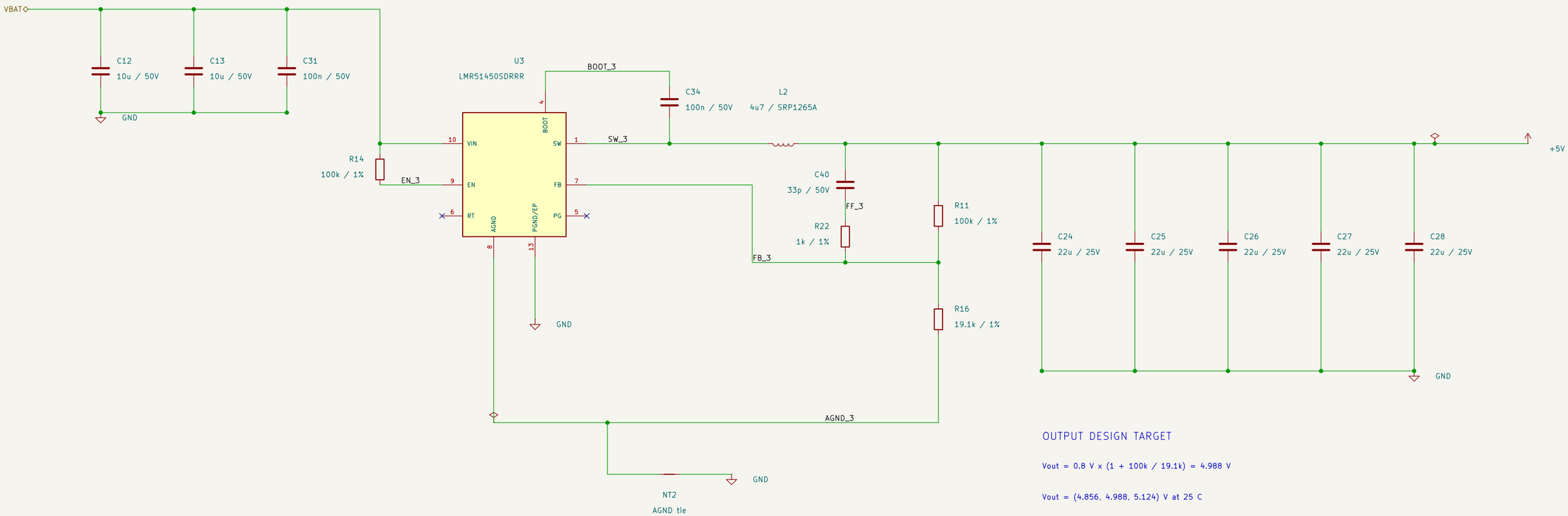
Size: A3	Date: 2026-09-30	**Rev: A6**
KiCad E.D.A. 10.0.6		Id: 3/7

5 V SERVO SUPPLY

VBAT = (19.8, 22.2, 25.2) V: planning battery range; no automatic cutoff.

INPUT AND ENABLE

REGULATOR / FILTER / FEEDBACK



OUTPUT DESIGN TARGET

Vout = 0.8 V x (1 + 100k / 19.1k) = 4.988 V

Vout = (4.856, 4.988, 5.124) V at 25 C

Bounds exclude ripple, transients and dropout.

Planning load: 2.4 A; board thermal rating unverified.

CONFIGURATION AND IMPLEMENTATION

LMR51450SDRRR / C6890985: 5 A IC, PFM + spread spectrum; RT open = 500 kHz; PG unused.

Input HF capacitor and BOOT capacitor: short loops at IC. Join AGND to PGND/EP13 locally through the copper net tie.

Stacked package pins: VIN 10–12; SW 1–3. Solder EP13. Feed–forward RC and effective output capacitance require validation.

EN is pulled to VBAT. No battery undervoltage cutoff. Verify loaded startup, load steps and temperature rise.

Source: TI SLUSEP7, sections 8–9 and DRR drawing; www.ti.com/lit/ds/symlink/lmr51450.pdf

A6: L2 stays 4.7uH / C780205, matching TI Table 9–2 (5V, 500kHz).

C24–C28: 5 x 22u / 25V; about 87.5u effective at 5V (typical). TI table: 66u after derating.

Reference–based count with margin. C29 removed; C2 bulk retained. Verify actual ripple, load steps and startup.

HUMAN REVIEW – not approved for fabrication.
Schematic only. Owner edits are authoritative.

KSRC Lunar Rover

Sheet: /5 V servo supply/
File: buck–5v.kicad_sch

Title: ROVER PDB | 5 V servo supply

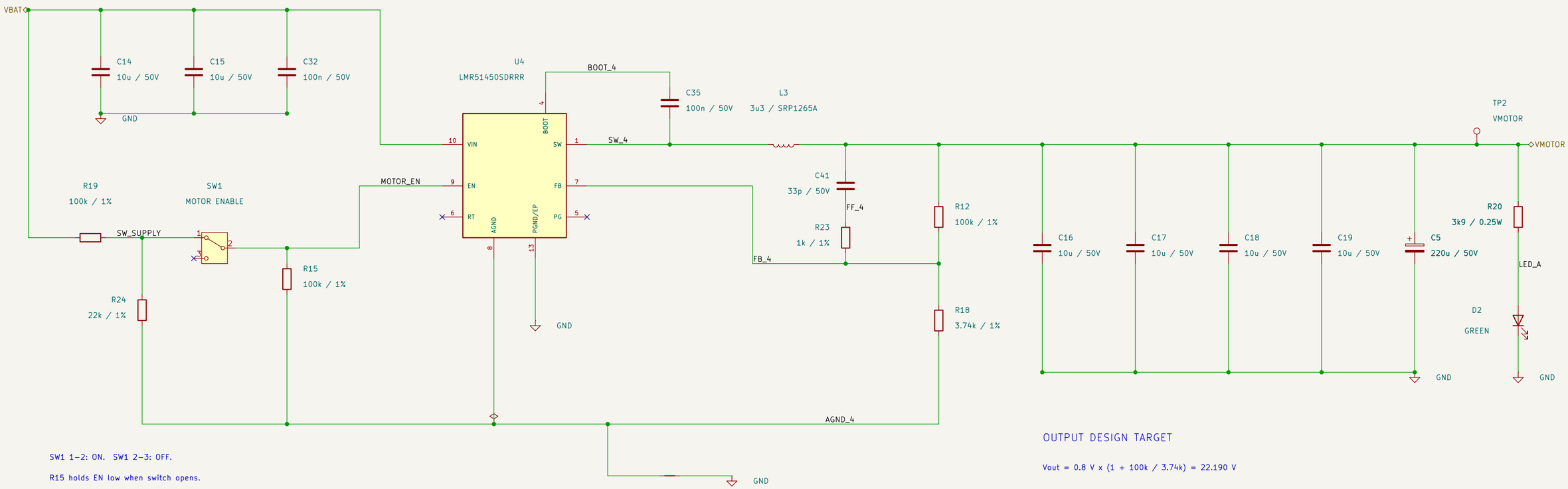
Size: A3	Date: 2026–09–30	**Rev: A6**
KiCad E.D.A. 10.0.6		Id: 4/7

22.19 V MOTOR SUPPLY

VBAT = (19.8, 22.2, 25.2) V: planning battery range; no automatic cutoff.

INPUT AND ENABLE

REGULATOR / FILTER / FEEDBACK



SW1 1–2: ON. SW1 2–3: OFF.

R15 holds EN low when switch opens.

ON: EN = 0.15278 x VBAT; OFF: ~0 V.

VIN remains connected to VBAT.

R19/R24 limit switch voltage; no +5V drive.

OUTPUT DESIGN TARGET

Vout = 0.8 V x (1 + 100k / 3.74k) = 22.190 V

Vout = (21.549, 22.190, 22.849) V at 25 C

Bounds exclude ripple, transients and dropout.

Planning load: 3.5 A; board thermal rating unverified.

D2: motor rail present; ~5 mA. Not a zero-voltage indication.

CONFIGURATION AND IMPLEMENTATION

LMR51450SDRRR / C6890985: 5 A IC, PFM + spread spectrum; RT open = 500 kHz; PG unused.

Input HF capacitor and BOOT capacitor: short loops at IC. Join AGND to PGND/EP13 locally through the copper net tie.

Stacked package pins: VIN 10–12; SW 1–3. Solder EP13. Feed-forward RC and effective output capacitance require validation.

SW1 is an enable command, not isolation. Rail drops with low battery; no braking clamp or active discharge.

Source: TI SLUSEP7, sections 8–9 and DRR drawing; www.ti.com/lit/ds/symlink/lmr51450.pdf

A7: L3 = 3.3uH / C1329348; ripple about 1.61A pk-pk at 25.2V in, 22.19V out, 500kHz.

C16–C19: retain 4 x 10u / 50V local ceramics. C5: retain 220u / 50V low-impedance bulk.

Input: retain 2 x 10u / 50V + 100n / 50V. DC-bias loss matters; nominal 20u is not effective 20u.

Basis: MOTOR_FILTER_SELECTION.md. Verify loaded startup, loop response, transients and dropout.

HUMAN REVIEW – not approved for fabrication.
Schematic only. Owner edits are authoritative.

KSRC Lunar Rover

Sheet: /22.19 V motor supply/
File: buck-motor.kicad_sch

Title: ROVER PDB \| 22.19 V motor supply

Size: A3	Date: 2026-10-01	**Rev: A9**
KiCad E.D.A. 10.0.6		Id: 5/7

PICO 2, SERVO OUTPUTS AND POWER CONNECTORS

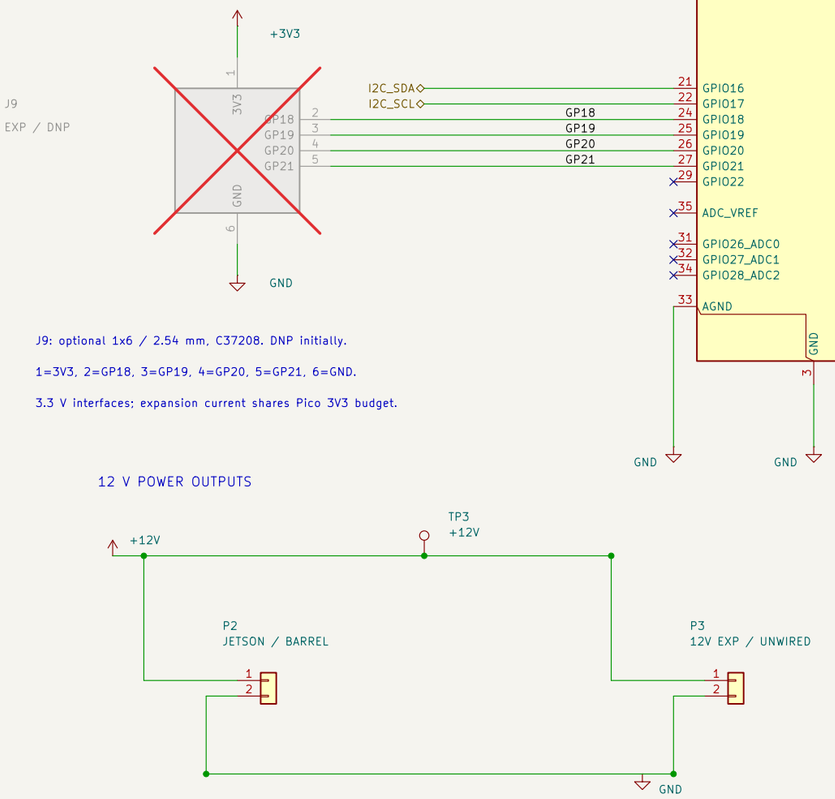
PICO POWER / EXPANSION

USB from Jetson provides power and data.

39 VSYS / 40 VBUS: unused on this PDB.

3V3 is Pico output, never Jetson header 3V3.

I2C to INA226 (sheet 2)



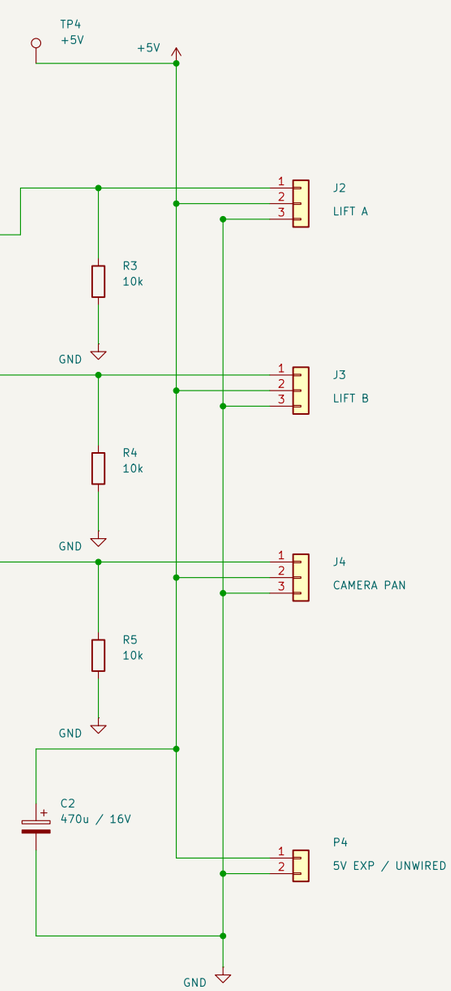
J9: optional 1x6 / 2.54 mm, C37208. DNP initially.

1=3V3, 2=GP18, 3=GP19, 4=GP20, 5=GP21, 6=GND.

3.3 V interfaces; expansion current shares Pico 3V3 budget.

SERVO CONNECTORS

PWM: Pico -> servos; 3.3 V logic



+5V source: sheet 4, 4.988 V nominal

2.4 A combined provisional servo load.

C2 is shared bulk capacitance.

P4 expansion initially unwired.

12 V POWER OUTPUTS

+12V source: sheet 3, 11.989 V nominal.

P2 powers Jetson; P3 expansion initially unwired.

J2–J4: pin 1 = PWM, pin 2 = +5V, pin 3 = GND. Verify actual servo pinout and missing-signal behavior.

C2 supports the shared servo rail. Place the three pull-downs near their PWM connections on the PCB.

SKT1 represents Pico 2 with two 1x20 sockets. All eight ground pins connected; debug pads not socketed. Keep USB/BOOTSEL accessible.

Pico 2 datasheet 5.4 / Figure 7: USB supplies the onboard VSYS regulator. No external supply is fitted.

USB power absent: Pico, INA226 and J9 3V3 are off. Servo power can remain on; verify missing-PWM behavior.

Motor enable remains independent of USB. Firmware must handle lost data while USB power is still present.

HUMAN REVIEW – not approved for fabrication.
Schematic only. Owner edits are authoritative.
KSRC Lunar Rover

Sheet: /Pico 2 and hobby servos/
File: pico-servos.kicad_sch

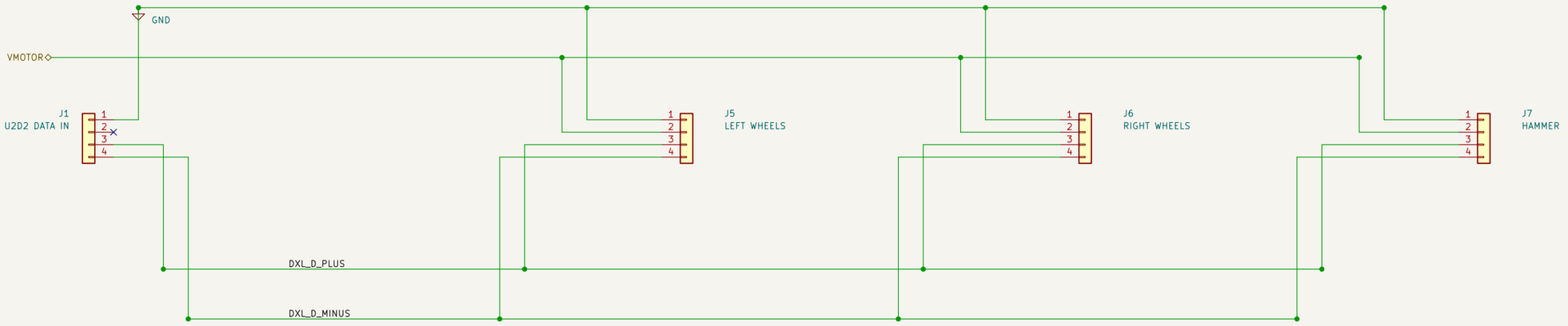
Title: ROVER PDB | Pico 2 and hobby servos

Size: A3	Date: 2026-10-01	**Rev: A9**
KiCad E.D.A. 10.0.6		Id: 6/7

DYNAMIXEL POWER AND DATA DISTRIBUTION

VMOTOR from sheet 5: 22.190 V target; lower in dropout.

D+/D− are a shared bidirectional bus. Connector drawings retain the physical pin order.



J1 pin 2 is intentionally unconnected; J5/J6/J7 pin 2 receives VMOTOR. One common D+/D− bus; no on-board transceiver or termination.

Motor switch, rail indicator and VMOTOR test point: sheet 5. No remote-stop jumper or connector.

A shorted output branch has no individual fuse. Review wiring/contact protection against converter fault current.

HUMAN REVIEW — not approved for fabrication.
Schematic only. Owner edits are authoritative.

KSRC Lunar Rover

Sheet: /DYNAMIXEL distribution/
File: dxl-enable.kicad_sch

Title: ROVER PDB / DYNAMIXEL distribution

Size: A3	Date: 2026-09-30	**Rev: A8**
KiCad E.D.A. 10.0.6		Id: 7/7